

Lecture 3: Statistics of stationary time series

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In Lecture 1 we reviewed how to decompose time-series as the combination of a trend, its cyclic components and a residual time-series. In turn, the residual series may be considered as a second order stationary time-series, with its trends and cycles — sometimes non-visible. This introduces the possibility of describing it as a second order stationary model to that residual time-series. In the following chapters we will review different second order stationary models. Our analysis of any time-series is complete when we prove that the residual-time series are IID noise according to a given law.

We previously saw that a second order stationary process is determined by the pair (μ, γ) or the triplet (μ, σ^2, ρ) . μ is the constant mean, γ the autocovariance function, which is symmetric and depends on the lag alone. σ^2 is the variance, which equals $\gamma(0)$. ρ is the autocorrelation function, which is also symmetric, $\rho(0) = 1$ and depends on the lag alone. Given any time-series $x_1, \dots, x_n$, assumed to be stationary, we need methods to estimate μ , σ^2 , γ and ρ .

1 Basic estimators of the characteristics of a second order stationary time series

Consider $x_1, \dots, x_n$, observed data from a stationary time series. To fit a stationary model we need to estimate the mean μ and the covariance function γ , or alternatively, to estimate the mean μ , the variance σ^2 and the autocorrelation function ρ . The usual estimators are listed next:

1. The empirical mean:

$$\bar{x}_n := \frac{1}{n} \sum_{i=1}^n x_i$$

2. The empirical variance:

$$\hat{\sigma}^2 = \hat{\gamma}(0) := \frac{1}{n} \sum_{j=1}^n (x_j - \bar{x}_n)^2$$

3. The empirical auto-covariance:

$$\hat{\gamma}(l) := \frac{1}{n} \sum_{j=1}^{n-l} (x_j - \bar{x}_n)(x_{j+l} - \bar{x}_n), \quad l < n$$

4. The empirical autocorrelation:

$$\hat{\rho}(l) := \frac{\hat{\gamma}(l)}{\hat{\gamma}(0)}$$

Note that a depiction of the empirical autocorrelation may be used as the basic tool to detect the best fitting model for our dataset. Given a model, we have the theoretical shape of its autocorrelation function and so, to identify a model that fits our data consist in to search a model with a theoretical shape of its autocorrelation function similar to the shape of the observed correlogram of our data. We will see how to do this with specific models in the following weeks.

2 Testing the IID noise hypothesis

The characterization of a time series ends when describing its residuals as IID noise. Recall that an IID noise is nothing but a collection of independent and identically distributed random variables, or in other words, an instance of any given probability law.

So, to have at hand a method to determine if a series of residuals is or not an IID noise is of major importance. The following result solves the problem; the proof can be found in [1].

Theorem 2.1 *Let $y_1, y_2, \dots, y_n$ be n observations of an IID noise. Let*

$$\{\hat{\rho}(j), 1 \leq j \leq h\}$$

be the graph of its autocorrelation function until the lag h . Then, for n big enough and h relatively small, values $\hat{\rho}(j)$ can be considered as a random sample of a $N(0, \frac{1}{n})$.

So, if the IID hypothesis holds true, 95% of the h observed autocorrelations have to lay within the interval

$$\left[-1.96 \frac{1}{\sqrt{n}}, 1.96 \frac{1}{\sqrt{n}} \right].$$

This is useful to test whether a given time-series is an IID noise. For example, given the first 50 values of $\hat{\rho}$, if 3 or more of them appear outside of this interval, we cannot consider the series as IID noise.

Remark 2.2 *If we have a time series with length n it is commonly accepted that values of $\hat{\rho}$ up to lag of order $\frac{n}{3}$ are significant. But some researchers consider only acceptable autocorrelations until lag $\log n$.*

A more precise method based on the previous idea in order to test if a time series is an IID noise is the Ljung-Box test (1978). It is based on the statistic

$$Q_{LB} := n(n+2) \sum_{j=1}^h \frac{\hat{\rho}^2(j)}{n-j}$$

where h is a parameter that has to be determined and $Q_{LB}(h) \sim \chi_h^2$. Recall that χ_h^2 denotes the chi-square distribution with h degrees of freedom.

If Q_{LB} is big, we reject IID noise hypothesis. If we have obtained $Q_{LB} = q$,

$$p := P\{Q_{LB} > q\} = P\{\chi_h^2 > q\}$$

is the p -value or risk to reject the hypothesis when it is true. So, usually, if $p < 0.05$, we reject the IID character of the series.

3 Testing Gaussianity

If we know that our residuals are an IID noise it can be interesting to test if it is in fact a Gaussian white noise. The traditional way to do this is using the well-known Kolmogorov-Smirnov goodness of fit test.

An alternative is the so called $Q - Q$ -plot:

Let $Y_1, \dots, Y_k$ be a sample of a $N(\mu, \sigma^2)$ law and let $X_1, \dots, X_n$ be a sample of a $N(0, 1)$ law. Let $X_{(1)}, \dots, X_{(k)}$ and $Y_{(1)}, \dots, Y_{(k)}$ the corresponding order statistics. Denote by $x_{(i)}$ and $y_{(i)}$ the corresponding empirical observations.

We have of course

$$\mathbb{E}(Y_{(j)}) = \mu + \sigma \mathbb{E}(X_{(j)}).$$

If $m_i := \mathbb{E}(X_{(i)})$, we can draw the pairs

$$(m_i, y_{(i)})$$

and the graphics of these pairs should be approximately linear. And in particular, its correlation should be near to 1. This is the basis of the so-called Shapiro-Wilk test (1965).

The value m_i is usually approximated by

$$m_i \sim \Phi^{-1}\left(\frac{i - 0.5}{n}\right)$$

where Φ is the cumulative probability of the standard normal distribution.

Therefore, we have

$$R^2 := \frac{(\sum (Y_{(i)} - \bar{Y}_n) \Phi^{-1}(\frac{i-0.5}{n}))^2}{\sum (Y_{(i)} - \bar{Y}_n)^2 \cdot \sum (\Phi^{-1}(\frac{i-0.5}{n}))^2} \in [0, 1],$$

and the p -value is the probability

$$P\{R^2 < r^2\}.$$

where r is the observed correlation.

The distribution of R^2 is the Shapiro-Wilk distribution. Its values can be found at

<http://www.real-statistics.com/statistics-tables/shapiro-wilk-table/>.

For $n \geq 50$, for example, we have

$$P\{R^2 < 0.987\} = 0.05.$$

So, if the observed r^2 is less than 0.987, we reject the hypothesis of gaussianity.

References

- [1] P. J. Brockwell and R. A. Davis (1991): *Time Series: Theory and Methods*. Springer.